

Resume Tailor Engine: Architecture & Cloud Deployment

Complete End-to-End System Documentation, Guardrail Design, and Streamlit Cloud Deployment Guide

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1. Executive Summary

The Resume Tailor system is an enterprise-grade, deterministic resume customization and technical interview preparation engine. Built specifically for high-throughput technical recruitment (Data Engineering, Big Data, Cloud Architecture), the system bridges the gap between applicant tracking systems (ATS) and human technical screeners by solving three fundamental problems:

- **Anti-Hallucination Guarantee:** Unlike standard LLM-based resume builders that invent false experiences, tools, or dates, this system strictly isolates the candidate's career data inside an immutable Source of Truth (SOT). Only verified credentials and production metrics are permitted.
- **Strict 1-Page Layout Engineering:** Human recruiters spend 6 to 8 seconds scanning resumes. The auto-fitting typography engine calculates layout geometry and guarantees every generated PDF fits on strictly 1 single page without awkward bottom whitespace or two-page spillovers.
- **24/7 Mobile Cloud Access:** Deployed to Streamlit Community Cloud (connected to GitHub), the platform allows the candidate to copy any Job Description (JD) on a smartphone, generate a tailored ATS resume in 5 seconds, download the PDF, and apply directly via mobile portals with zero laptop dependency.

2. Core Architecture & Pipeline Components

Pipeline Flow: Job Description (JD) → NLP Keyword Extraction & SOT Matcher → Config Synthesis → Deterministic ATS Validator → Auto-Fitting Word/PDF Renderer → 7-Day Technical Interview Roadmap → Tracker & Cloud Sync.

A. Source of Truth (SOT) Data Layer

Located in the `source_of_truth/` directory, this layer acts as the single source of verified career records:

- **profile.yaml:** Stores verified candidate identity, contact information, employer records (TCS / State Bank of India), client projects (Vendors Payment & Settlement System), and official titles.
- **skills.yaml:** Exhaustive taxonomy of technical proficiencies across 8 distinct categories (Big Data, Languages, Cloud & Warehouses, Orchestration, Databases, Data Quality, Modeling, Tools) tagged with certification associations.
- **certs_projects.yaml:** Maintains real verified certifications (Databricks Data Engineer Associate, AWS Solutions Architect Associate, AWS Cloud Practitioner, and Microsoft Certified: Fabric Data Engineer Associate ID: F2CEA18F28A701DC) and production projects.
- **bullets_bank.yaml:** Pre-approved STAR-format metric bullets (Situation, Task, Action, Result) capturing real impact: 4.45M+ daily transaction hits, 12-hour settlement cycles, 30% pipeline downtime cut, and 99.8% SLA adherence.

B. Deterministic ATS Validation Engine (`scripts/validate_resume.py`)

Before any resume is exported, an automated validator executes strict integrity assertions:

- **SOT Keyword Whitelist:** Every single bullet, company, and skill is matched against the SOT. If an unapproved keyword or fabricated metric appears, validation fails immediately.
- **Certification Verifiers:** Ensures credential IDs, issuance dates, and expiration dates match official certifying authority records.

- **Banned Buzzwords Guardrail:** Rejects non-actionable fluff phrases (e.g. 'results-driven team player') in favor of concrete engineering terminology.

3. Auto-Fitting Typography & Single-Page Engineering

One of the most technically complex challenges in automated resume generation is ensuring the output fills exactly one page regardless of bullet variations. Our custom rendering engine in `scripts/render_resume.py` implements several innovations:

- **Native Right-Aligned Tab Stops:** Replaced brittle space-padded alignment with native Word tab stops (`WD_TAB_ALIGNMENT.RIGHT` at `Inches(7.27)` with tab characters). Dates and locations remain perfectly flush against the right margin across all devices.
- **Dual-Profile Typography Engine:** Includes 'spacious' (19pt name, 10pt headers, 9.0pt body, 1.08-1.10 line spacing) and 'compact' (17pt name, 9.5pt headers, 8.5pt body, 1.02 line spacing) calibrated profiles.
- **Automated Page-Count Feedback Loop:** After rendering, `pypdf` programmatically inspects the output PDF. If page count > 1, the engine automatically rolls back, selects the compact profile, and re-renders to guarantee a strict 1-page fit.
- **Cross-Platform Converter (`convert_to_pdf`):** Uses Microsoft Word COM automation via `docx2pdf` on Windows local machines, and automatically falls back to headless LibreOffice (`libreoffice --headless --convert-to pdf`) on Linux cloud environments.

4. Dedicated 7-Day Technical Interview Roadmap Engine

Generating a resume is only half the battle; the candidate must clear the technical interview. Every tailoring execution automatically generates a `roadmap.md` inside the job directory containing:

- **Day 1 - Spark Architecture:** Deep dive into Catalyst Optimizer, Tungsten execution engine, DAG scheduling, Stages, and Tasks.
- **Day 2 - PySpark DataFrame API:** Transformation patterns, avoidance of Python UDFs, broadcast joins, and partitioned data loading.
- **Day 3 - Performance Tuning & OOM Debugging:** The core SBI production story: diagnosing Out-Of-Memory exceptions, tuning `spark.sql.shuffle.partitions`, and eliminating data skew.
- **Day 4 - Advanced SQL Transformations:** Analytical window functions (`ROW_NUMBER`, `DENSE_RANK`, `LEAD/LAG`), CTEs, and Oracle PL/SQL optimization.
- **Day 5 - Orchestration & Resiliency:** Apache Airflow DAG design, idempotency, failure retries, and SLA alerting.
- **Day 6 - Cloud Lakehouse & Delta Architecture:** Databricks & Microsoft Fabric, ACID transaction log, time travel, and Z-ORDER indexing.
- **Day 7 - Company Mock Interview:** STAR-format project walkthrough and company-specific question drills.

5. 100% Free 24/7 Cloud Deployment on Streamlit Cloud

To eliminate laptop dependency when applying on mobile, the system is deployed to Streamlit Community Cloud (backed by Snowflake) connected directly to GitHub (`wadekarhushikesh06/JD_cust_hrushi`):

Deployment Attribute	Implementation Detail
Cloud Platform	Streamlit Community Cloud (Free Tier, 24/7 Hosting, Zero Cost Forever)
Billing / Card Required?	None. 100% Free for GitHub developers (Zero credit/debit card needed)
Linux Headless Packages	<code>packages.txt</code> specifies <code>libreoffice</code> , <code>fonts-liberation</code> , <code>fonts-dejavu</code> for cloud PDF rendering

Application Entry Point

app.py provides a touch-friendly mobile interface with 1-click PDF download & roadmap preview

6. Universal Setup & Replication for Any Candidate

To configure this exact platform for a new candidate or different machine, follow this 4-step sequence:

- **1. Clone Repository:** `git clone https://github.com/wadekarhrushikesh06/JD_cust_hrushi.git` & `cd JD_cust_hrushi`
- **2. Install Dependencies:** `pip install -r requirements.txt`
- **3. Run Interactive Setup:** `python scripts/setup_new_user.py` (prompts for candidate name, contact, employer, and title)
- **4. Deploy to Streamlit Cloud:** Push to personal GitHub repository, open share.streamlit.io, click 'Create app', and select app.py.